

E03 Equilibrium limited PFR example

The elementary reaction $A \rightleftharpoons H_2 + C$ is carried out isothermally and isobarically in a PFR. The feed is pure A, with $C_{A0} = 0.2 \text{ M}$ and $F_{A0} = 10 \text{ mol/min}$. The forward rate constant $k_f = 0.7 \text{ min}^{-1}$ and the equilibrium constant $K_C = 0.05 \text{ M}$. The PFR has a volume of 500 L. Determine the conversion for the PFR.

For review, let's solve the PFR part of the example again, using two methods:

- First, kinetically:

- Now let's solve for the equilibrium conversion, i.e., thermodynamically ... To do that, we don't need to solve any reactor design equations:

Decision Guide: PFR Example Solved Two More Ways

Goal

Determine:

1. The conversion in the PFR.
2. The equilibrium conversion.
3. How a conversion-based approach differs from an F-based approach.

Step 1: Identify What Kind of Problem This Is

Ask yourself:

What reactor?

→ PFR

Is it isothermal?

→ Yes

Is it isobaric?

→ Yes

Single reaction?

→ Yes

Because the problem can be expressed using a single independent conversion variable, either:

- Species mole balances (F form), or
- Conversion-based balances (X form)

can be used.

Step 2: Write the PFR Mole Balance

Use the conversion form of the PFR mole balance:

$$F_{A0} \frac{dX}{dV} = -r_A$$

CBE 3610 Reactor Design Handbook Equation (22)

This is the starting point for the conversion-based solution.

Step 3: Write the Rate Law

The reaction is:

Elementary

Reversible

Start with:

$$-r_A = [k(T)] \cdot [f(C_A, C_B)]$$

CBE 3610 Reactor Design Handbook Equation (44)

Then incorporate reversibility using the equilibrium relationship provided in the problem statement.

Step 4: Express Concentrations in Terms of Conversion

Because the system is:

Gas phase

Isothermal

Isobaric

use the concentration-change relationship:

$$C_j = C_{A0} \frac{\Theta_j + \nu_j X}{1 + \epsilon X} \left(\frac{P}{P_0} \right) \left(\frac{T_0}{T} \right)$$

CBE 3610 Reactor Design Handbook Equation (55)

For this problem:

Pure A feed, so many terms simplify considerably.

Step 5: Substitute Into the Mole Balance

Insert:

- Rate law
- Concentration expressions

into Equation (22).

This creates:

$\frac{dX}{dV} = f(X)$ which can be integrated from: $X = 0$ to $X = X_{exit}$ over the reactor volume.

Step 6: Recognize That Equilibrium Conversion Is Different

The second part of the example asks for:

Equilibrium Conversion

This is **not** a reactor-design calculation.

No mole balance is required.

No integration is required.

No reactor volume is required.

Step 7: Write the Equilibrium Expression

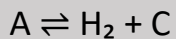
Use the equilibrium constant relationship:

$$K_C = \frac{C_C C_{H_2}}{C_A}$$

CBE 3610 Reactor Design Handbook Equation (66)

Step 8: Express All Concentrations in Terms of Conversion

Using stoichiometry:



write:

$$C_A(X) \\ C_{H_2}(X) \\ C_C(X)$$

using Equation (55).

Step 9: Solve for Equilibrium Conversion

Substitute the conversion expressions into Equation (66).

This reduces the equilibrium condition to:

$$K_C = f(X_e)$$

Solve for: X_e

No reactor-design equation is needed.

Step 10: Compare the Two Approaches

Reactor Design Calculation

Uses:

Rate Law

+

Mole Balance

to determine conversion in a finite reactor.

Equilibrium Calculation

Uses:

Stoichiometry

+

Equilibrium Constant

to determine the maximum conversion thermodynamically possible.

Final Answer Checklist

Before moving on, verify that you have:

- ☒ Written the PFR mole balance using Equation (22)
- ☒ Expressed concentrations using Equation (55)
- ☒ Written the reversible rate law using Equation (44)
- ☒ Integrated the PFR design equation to determine reactor conversion
- ☒ Written the equilibrium expression using Equation (66)
- ☒ Solved separately for equilibrium conversion

CBE 3610 Reactor Design Handbook Equations Needed

Eq. (22) PFR Mole Balance

Eq. (44) General Rate Law

Eq. (55) Concentration as a Function of Conversion

Eq. (66) Equilibrium Constant in Terms of Concentration

Key Insight

Reactor Conversion



Requires Rate Law + Mole Balance

Equilibrium Conversion



Requires Stoichiometry + KC

One calculation determines **what a reactor achieves**, while the other determines **what thermodynamics allows**.